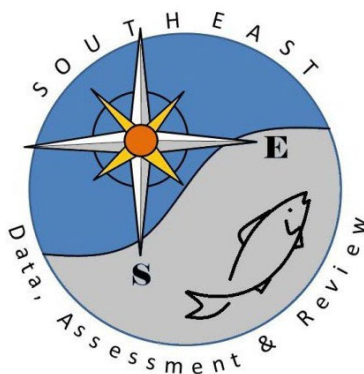




SEDAR

Southeast Data, Assessment, and Review

SEDAR 84

US Caribbean Stoplight Parrotfish and Yellowtail Snapper

SECTION IV: Research Recommendations

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1. DATA WORKSHOP RESEARCH RECOMMENDATIONS

1.1 LIFE HISTORY RESEARCH RECOMMENDATIONS

Issue 1: Are sufficient life history data available?

- Life history sampling should be done with statisticians to ensure more efficient collection programs (e.g., optimizing sample sizes within size bins).
- Ensure statistically robust sample sizes of small and large size classes of fish.

1.2 COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

1.2.1 Commercial Landings Research Recommendations

Issue 1: Are analysis-ready commercial landings data available for SEDAR 84?

- Investigate trends in effort, significant socioeconomic and environmental events, and associated effects on the demographics, gears used, and species landed.
- Increased port sampling is needed in St. Croix to enable analyses required for quantifying removals.
- Conduct recreational fishery port sampling surveys to determine removals due to recreational fishing.
- Investigate the applicability of hindcasting for all parrotfish combined or other applicable future assessments.

Issue 2: Should data outliers in the commercial landings be flagged for additional investigation?

- Operationalize an outlier flagging process for future SEDAR assessments.

Issue 3: What should the gear fleets be for the commercial landings data?

- Operationalize a gear grouping process for future SEDAR assessments.

1.2.2 Length Composition Research Recommendations

Issue 1: Are analysis-ready size data available for SEDAR 84?

- Increase collection efforts to increase sample size in TIP.
- Work with port samplers and fishers to implement the trip interview program better and ensure critical fishing times are captured adequately (e.g., at night).
- Operationalize an outlier flagging process for future SEDAR assessments.
- Develop a data management system to link TIP to CCL. A linked system would require changing the overarching structure of collecting fishery-dependent data. A fishery information network system for the US Caribbean could comprehensively resolve this

issue.

- Investigate if relative weight at a given length has changed across years or clusters of years.
- Examine the number of trips when considering the representativeness of TIP samples; do not limit the investigation to the number of fish or the weight of the catch, samples, and trips.
- Going forward, we need a recommendation on how to do a new sampling strategy that is more holistic.

1.2.3 Discards and Discard Mortality Research Recommendations

Issue 1: Do we have estimates of commercial discards and estimates of discard mortality?

- As shark depredation could play an increasing role in discard mortality, additional research is needed to quantify discards better.
- Promote, through outreach and education, increased reporting of discards.

1.3 RECREATIONAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

No research recommendations were provided.

1.4 MEASURES OF POPULATION ABUNDANCE RESEARCH RECOMMENDATIONS

Issue 1: Should the fishery-independent density estimates from NCRMP be used in SEDAR 84?

- Use the NCRMP data to investigate what has occurred in regions consistently sampled since 2001 (e.g., Buck Island). Doing so may require combining data across years.
- Investigate the potential impact of changes in habitat on the surveys.
- There is an association between the habitat and fish, but we should consider whether the habitat changes the spatial distribution of fish.
- Investigate highly turbid areas that are currently not surveyed.
- Expand fishery-independent surveys to seagrass/mangrove habitats since these areas are essential for recruitment.

Issue 2: Should the fishery-dependent information from commercial logbooks be considered to conduct abundance indices?

- For future SEDARs, consider an ecosystem-based perspective by investigating what species groups and associated species commercial fishers target to understand species complexes better.
- For future assessments, filter out trips that report only lobster or conch to understand the targeted effort better.
- Investigate the relationship between the catch and effort of the diving data to document the disconnect between time diving and species-specific targeted effort for species

considered bycatch or opportunistically targeted.

2. ASSESSMENT PROCESS RESEARCH RECOMMENDATIONS

To mitigate some of the data uncertainties it is recommended to:

- Expand fishery-independent survey time series and resolution (e.g., retain and use 1-cm length bin data where available).
- Further evaluate natural mortality and growth assumptions. Collect and analyze additional life history data to reduce uncertainty around growth and natural mortality rates.
- Conduct focused research on historical catches and fishing history to inform and constrain early model conditions.
- Consider using simpler production models or age-structured models with fixed selectivity to isolate and evaluate different data inputs.
- Develop and evaluate model ensembles or uncertainty grids to guide catch advice under different plausible scenarios.
- Investigate stock connectivity to better understand local versus regional recruitment dynamics and their implications for informing steepness.
- Re-examine multiple indicator approach for the Parrotfish Two management unit given that SEDAR 84 only comprehensively evaluated data for *Sparisoma viride*.
- Research methods, including simulations, to “right-size” model complexity to match data availability, avoiding overparameterization in data-limited contexts.
- Support Management Strategy Evaluations that are robust to key uncertainties to guide harvest advice.
- Ensure the continuation of fishery-independent survey programs (e.g., National Coral Reef Monitoring Program) with consistent spatial and temporal coverage.
- Maintain and expand commercial catch monitoring programs. Expand port sampling and other fishery-dependent data collection to fill gaps in length composition and effort data.

3. REVIEW PANEL RESEARCH RECOMMENDATIONS

The RP supports the research recommendations outlined in the DW and AW reports for the U.S. Caribbean Stoplight Parrotfish assessment processes. Based on those recommendations, the following were identified as high priority and are listed in order of importance:

St. Croix Stoplight Parrotfish

- Short term
 - Investigate the applicability of hindcasting for all parrotfish combined or other applicable future assessments.
 - For example, the proportion of Stoplight Parrotfish could be investigated

- from 2012, and a mean of that proportion could be applied to split the catches prior to the species-specific split
 - Conduct focused research on historical catches and fishing history to inform and constrain early model conditions
- Further evaluate natural mortality and growth assumptions
 - Explore the use of natural mortality at age
- Investigate the relationship between the catch and effort of the diving data to document the disconnect between time diving and species-specific targeted effort for species considered bycatch or opportunistically targeted.
- Investigate stock connectivity to better understand local versus regional recruitment dynamics and their implications for informing steepness
- Long term
 - Ensure statistically robust sample sizes of small and large size classes of fish
 - Maintain and expand commercial catch monitoring programs. Expand port sampling and other fishery-dependent data collection to fill gaps in length composition and effort data.
 - Increase collection efforts to increase sample size in TIP
 - There are several biases in the current NCRMP survey design. Notably, the survey is conducted over a limited time frame (for two weeks every two years), which would miss seasonal or environmental changes. Additionally, the survey may not be adequately capturing the full size distribution of the population (e.g., upwelling/high turbulence areas that have high density of larger fish, underrepresentation of young of the year etc.). The RP support the following recommendations for the NCRMP survey:
 - Ensure the continuation of fishery-independent survey programs (e.g., National Coral Reef Monitoring Program) with consistent spatial and temporal coverage
 - Expand fishery-independent survey time series and resolution
 - Investigate the potential impact of changes in habitat on the surveys.
 - There is an association between the habitat and fish, but we should consider whether the habitat changes the spatial distribution of fish.
 - Investigate highly turbid areas that are currently not surveyed.
 - Expand fishery-independent surveys to seagrass/mangrove habitats since these areas are essential for recruitment.
 - Conduct recreational fishery port sampling surveys to determine removals due to recreational fishing

The RP also recommends the following:

- St. Croix Stoplight Parrotfish
 - Conduct population structure studies through genetics, tagging, fish larvae, otolith microchemistry, or modelling.
 - Improve otolith samples

St. Thomas/ St. John and Puerto Rico Yellowtail Snapper

The RP supports the research recommendations outlined in the DW and AW reports for the U.S. Caribbean Yellowtail Snapper assessment processes. Based on those recommendations, the following were identified as high priority and are listed in order of importance:

Short term

- Investigate the applicability of hindcasting for all parrotfish combined or other applicable future assessments (for St. Thomas/St. John commercial catch data)
- Further evaluate natural mortality and growth assumptions
 - Explore the use of natural mortality at age
- Explore parameterizing retention to improve selectivity of the commercial fleet and interpret the apparent high selectivity of larger individuals that are poorly estimated by the current models.
- Request for USVI fisher to provide their logbooks to further investigate the fisher behavior as an index assignment for CPUE.
- Investigate stock connectivity to better understand local versus regional recruitment dynamics and their implications for informing steepness

Long term

- There are several biases in the current NCRMP survey design. Notably, the survey is conducted over a limited time frame (for two weeks every two years), which would miss seasonal or environmental changes. Additionally, the survey may not be adequately capturing the full size distribution of the population (e.g., upwelling/high turbulence areas that have high density of larger fish, underrepresentation of young of the year etc.). Currently, there is limited DCRMP data, which would target larger, older fish that are in deeper waters and help complement the NCRMP survey. The RP support the following recommendations for the NCRMP and DCRMP surveys:
- Ensure the continuation of fishery-independent survey programs (e.g., National Coral Reef Monitoring Program) with consistent spatial and temporal coverage
- Expand fishery-independent survey time series and resolution
- Continue DCRMP work in the future.
- Look into National Parks transect data and compare it to the NCRMP data
- Maintain and expand commercial catch monitoring programs. Expand port sampling and other fishery-dependent data collection to fill gaps in length composition and effort data

- Increased port sampling is needed in St. Thomas and St. John to enable analyses required or quantifying removals.
- Increase collection efforts to increase sample size in TIP.
- Conduct recreational fishery port sampling surveys to determine removals due to recreational fishing.

The RP also recommends the following:

- St. Thomas/St. John and Puerto Rico Yellowtail Snapper
 - Further investigate the population structure of those two stocks.
 - Conduct population structure studies through genetics, tagging, fish larvae, otolith microchemistry, or modelling.
 - Improve otolith samples
 - Reconsider not removing old NCRMP survey data (before 2013/2014; La Parguera)